Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad de Valladolid
Short name / acronym used in projects	UVA
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	www.ioba.es
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Paseo de Belén 17, 47011, Valladolid, SPAIN
Department(s), institute(s), research group(s) or unit(s) that would be involved	Department of Organic Chemistry, Organosilanes in Organic Synthesis group (OSOS group), University of Valladolid
Additional institutional resources relevant to collaboration	Facilities and equipment of OSOS: standard facilities of a lab of Organic Synthesis and centralized services of UVA (https://laboratoriotecnicasinstrumentales.es/).

Main contact person for this profile	Asunción Barbero Pérez, Full Professor of Organic Chemistry, Department of Organic Chemistry, University of Valladolid
Main contact email	asuncion.barbero@uva.es
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): Synthesis of drug analogs

Free Keywords for internal searchable database (please provide up to 10)	e.g. retinal organoids; ocular surface inflammation; zebrafish; AI diagnostics; drug delivery
Geographies of interest for exchanges (optional)	Germany, Ireland, UK, Poland, Italy

Overview of your research / innovation focus
Our research focuses on the design and synthesis of novel heterocyclic scaffolds with potential applications in medicinal chemistry and drug discovery. Our aim is the development of a broad range of chemically and structurally diverse heterocycles using innovative synthetic strategies, including Prins-type cyclizations. These compounds provide a versatile platform for exploring new molecular entities with potential therapeutic relevance. Previous studies in our group have highlighted the potential of several of our scaffolds to exert neuroprotective effects. In particular, preliminary results indicate that a compound synthesized in our laboratory, featuring a dioxaspirobicyclic scaffold, protects aging neurons against amyloid oligomer toxicity. These promising findings provide a basis for further investigating novel heterocycles as potential neuroprotective agents. We are also investigating silicon-containing heterocycles as potential modulators of disease-associated protein interactions. Our preliminary studies have demonstrated that three organosilicon compounds bind to hotspot regions of NUPR1, an intrinsically disordered protein associated with pancreatic ductal adenocarcinoma. Although these compounds exhibit moderate binding affinity, initial cellular studies have not revealed significant biological effects. These findings underscore the need to optimize ligand–protein interactions and explore additional chemical structures to improve biological activity. Within a translational eye-health and vision consortium, our expertise in synthetic and medicinal chemistry could contribute to the discovery of novel molecular scaffolds for investigating pathways involved in retinal degeneration, neuronal dysfunction, and other vision-related disorders. Our experience in developing structurally diverse heterocycles, combined with preliminary evidence of neuroprotective activity, provides a promising starting point for exploring their potential applications in ocular neuroprotection.

Why your team could be relevant to a translational eye-health and vision consortium
Our team could contribute to a translational eye-health and vision consortium through its expertise in the design, synthesis, and characterization of novel heterocyclic compounds with potential biomedical applications. These innovative molecular scaffolds may provide valuable starting points for the development of new therapeutic agents

targeting biological pathways relevant to ocular diseases, including retinal degeneration, glaucoma, and other vision-threatening conditions.
Our experience in medicinal chemistry and the development of structurally diverse molecular entities could complement the consortium's expertise in ophthalmology, vision science, disease modelling, and preclinical evaluation. In particular, the compounds and synthetic methodologies developed by our team could support the identification and optimization of new candidate molecules, provided that their biological activity, ocular relevance, and safety are established through appropriate experimental studies.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Our core expertise lies in the development and application of innovative synthetic methodologies for the construction of a wide variety of polysubstituted heterocycles with potential applications in medicinal chemistry and drug discovery. We can contribute expertise in molecular scaffold design, synthetic route development, and the preparation of structurally diverse chemical compounds that may serve as starting points for identifying new therapeutic candidates.
Our overarching goal is to bridge fundamental synthetic chemistry and biomedical research by generating innovative chemical entities whose biological potential in ocular diseases can be evaluated by our consortium partners.
Our approach is based on **flexible, collaborative research**, allowing our synthetic chemistry expertise to be adapted to the consortium's priorities and specific research questions. Through interdisciplinary collaboration, we aim to support early-stage drug discovery, facilitate the identification of novel chemical tools and potential therapeutic leads, and contribute to the development of innovative strategies addressing unmet needs in eye health and vision science.

Infrastructure, equipment, facilities or datasets relevant to consortium work

The group's laboratory, which is allocated at the department of Organic Chemistry (Faculty of Science), is well equipped with the required materials in a laboratory of Organic Synthesis (fume hoods, dry solvents systems, preparative and analytical HPLC, etc). The Faculty also provides facilities for all the research groups, such as stock rooms, chemical disposal, etc. Regarding the needed equipment for compound's characterization, the Scientific Park of the University has four NMR machines (two 400 and two 500 MHz), high resolution mass spectrometry (including MALDI-TOF, MS-TOF spectrometers), HPLC chromatography, X Ray machine, etc.

Training capacity and knowledge-transfer potential

Our team offers strong expertise in the design and synthesis of structurally diverse, polysubstituted heterocycles with potential applications in medicinal chemistry and drug discovery. We develop innovative synthetic methodologies, including Prins-type cyclizations, to access novel molecular scaffolds that can serve as starting points for identifying new bioactive compounds.
We are particularly interested in interdisciplinary collaboration, combining our synthetic chemistry capabilities with partners' expertise in biological screening, molecular modelling, and retinal disease research. Our flexibility in developing new chemical structures, together with our commitment to knowledge exchange, joint research, and researcher training, makes us a valuable partner for collaborative projects aimed at discovering innovative therapeutic approaches for eye health and vision disorders.

What your team would hope to gain from participating

Participating in the translational eye-health and vision consortium would provide our team with an excellent opportunity to expand our research towards ocular diseases and strengthen the connection between synthetic chemistry and biomedical applications. We hope to establish collaborations with experts in vision science and retinal disease modelling to evaluate the biological potential of our novel heterocyclic scaffolds and explore their possible applications in neuroprotection and the treatment of vision-related disorders.

We also aim to gain access to complementary expertise, advanced biological models, and innovative methodologies that would enable us to better understand the mechanisms of action of promising compounds and guide the design of new molecular entities with improved biological properties. The consortium would provide a valuable framework for exchanging knowledge, sharing resources, and developing interdisciplinary research projects that bridge fundamental chemical research and preclinical validation.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Asunción Barbero Pérez	Full Professor of Organic Chemistry	Organic Chemistry	Researcher	Experienced	asuncion.barbero@uva.es
Laura Fernández Peña	Assistant Professor	Organic Chemistry	Researcher	Experienced	laura.fernandez.pena@uva.es
Lucía García de la Peña	PhD Candidate	Organic Chemistry	Researcher	Early-Stage	lucia.garcia.parte@uva.es
Lucía Pedraza	PhD Candidate	Organic Chemistry	Researcher	Early-Stage	lucia.pedraza@uva.es

Main disciplines represented in your team	Organic synthesis, drug development, heterocyclic chemistry, synthesis of biologically active compounds, Structure determination of organic compounds
Experience in EU / international / multi-partner collaborative projects	Participation as an experienced researcher in two projects: H2020-NMP-2014-646075 and FP-Health-2011- 278557
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. Peña, L. F.; Estaras, M.; González-Andrés, P.; Díez-Poza, C.; Rizzuti, B.; Abian, O.; Velázquez-Campoy, A.; Iovanna, J. L.; Santofimia-Castaño, P.; Neira, J. L. Barbero, A. "Organosilicon molecules bind to the intrinsically disordered protein NUPR1 by clamping its hot-spots" Arch. Biochem. Biophys., 2025, 771, 110513. https://doi.org/10.1016/j.abb.2025.110513

2. Tapias, V.; González-Andrés, P.; Peña, L. F.; Barbero, A.; Núñez, L.; Villalobos, C. "Therapeutic Potential of Heterocyclic Compounds Targeting Mitochondrial Calcium Homeostasis and Signaling in Alzheimer's Disease and Parkinson's Disease" Antioxidants 2023, 12, 1282. https://doi.org/10.3390/antiox1206128

3. Díez-Poza, C.; Fernández-Peña, L.; González-Andrés, P. "Changing the Reaction Pathway of Silyl-Prins Cyclization by Switching the Lewis Acid: Application to the Synthesis of an Antinociceptive Compound" J. Org. Chem. 2023, 88, 6776−6783. https://doi.org/10.1021/acs.joc.3c00050

4. Peña, L. F.; Barbero, A. "Terminal cyclopropylsilyl alcohols as useful key units to access 2,3,4,6-tetrasubstituted tetrahydropyran scaffolds by stereocontrolled Prins cyclization" Org. Lett., 2024, 26, 5202–5207. https://doi.org/10.1021/acs.orglett.4c01806

5. P. González-Andrés, P.; Cherubin, A.; Barbero, A. "Expanding the silyl-Prins toolbox: selective access to C4-quaternary stereocenters in tetrahydropyrans using internal cyclopropylsilyl alcohols"Org. Chem. Front., 2025, 12, 6513–6518. 6518. https://doi.org/10.1039/d5qo00984g

Up to 5 relevant previous projects, grants, networks or collaborations

TITLE: "Bases celulares y moleculares del envejecimiento neuronal y de la susceptibilidad asociada a las enfermedades neurodegenerativas "

FINANCIAL ENTITY: Junta de Castilla y León, VA294P18
ROLE: COLLABORATOR
AMOUNT OF FUNDING: 120.000,00 €
FROM: 01-01-2019 TILL: 01-01-2021 NATIONAL

TITLE: Design and synthesis of bio-based polymers from sugar-derived monomers
FINANCIAL ENTITY: MICINN (MINISTERIO DE CIENCIA E INNOVACIÓN) TED2021-131705B-C21
ROLE: COORDINATOR
AMOUNT OF FUNDING: 109.940,00 €
FROM: 01-12-2022 TILL: 30-10-2025 NATIONAL

TITLE: Modulación de los canales de calcio operados por depósitos y señal de calcio en cáncer y envejecimiento neuronal
FINANCIAL ENTITY: MICINN (MINISTERIO DE CIENCIA E INNOVACIÓN) PID2021-125909OB-I00
ROLE: COLLABORATOR
AMOUNT OF FUNDING: 193.600,00 €
FROM: 01/09/2022 TILL: 31/08/2025

TITLE: "ELASTISLET-Tailored elastin-like recombinamers as advanced systems for cell therapies in diabetes mellitus: as synthetic biology approach towards a bioeffective and immunoisolated biosimilar islet/cell niche. Elastislet"
FINANCIAL ENTITY: European Commission, H2020-NMP-2014-646075
ROLE: COLLABORATOR
TOTAL AMOUNT OF FUNDING: 6.214.495,00 € AMOUNT OF FUNDING UVA: 1.629.250,00 €
FROM: 01-05-2015 TILL: 31-05-2019 INTERNATIONAL

TITLE: "THE GRAIL-Tissue in host engineering guided regeneration of arterial intimal layer"
FINANCIAL ENTITY: European Commission, FP-Health-2011- 278557
ROLE: COLLABORATOR
TOTAL AMOUNT OF FUNDING: 5.998.438,00 € AMOUNT OF FUNDING UVA: 787.800,00 €
FROM: 01-01-2012 TILL: 31-12-2016 INTERNATIONAL

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
None

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	X Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Our team offers strong expertise in the design and synthesis of structurally diverse, polysubstituted heterocycles with potential applications in medicinal chemistry and drug discovery. We develop innovative synthetic methodologies,

including Prins-type cyclizations, to access novel molecular scaffolds that can serve as starting points for identifying new bioactive compounds.
We are particularly interested in interdisciplinary collaboration, combining our synthetic chemistry capabilities with partners' expertise in biological screening, molecular modelling, and retinal disease research. Our flexibility in developing new chemical structures, together with our commitment to knowledge exchange, joint research, and researcher training, makes us a valuable partner for collaborative projects aimed at discovering innovative therapeutic approaches for eye health and vision disorders.

Capabilities or expertise you would especially like to access through the consortium
Through the consortium, we would particularly value access to specialized biological expertise and experimental platforms to evaluate the biological properties of our synthesized compounds and assess their potential activity in ocular disease models. We are also keen to develop expertise in molecular docking and computational modelling to investigate ligand–target interactions, guide compound prioritization, and support the rational design of new molecules with improved therapeutic potential. We would welcome opportunities to undertake short research stays and hands-on training with consortium partners to learn specific techniques in retinal biology and molecular docking. This knowledge transfer would allow us to establish new experimental collaborations, better interpret biological results, and integrate biological evidence into the design of our next generation of compounds. Another important benefit would be the development of joint research proposals targeting the intersection of synthetic chemistry, neuroprotection, and retinal disease. By combining our expertise in heterocyclic synthesis with partners' capabilities in vision science, we could generate preliminary data for competitive national and international funding applications, including collaborative projects focused on identifying new chemical candidates for vision-threatening diseases. Finally, we aim to develop lasting partnerships through joint publications, researcher exchanges, shared experimental protocols, and coordinated dissemination activities.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: ASUNCIÓN BARBERO PÉREZ
Date completed: September the 16th

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)